REV	DATE	DESCRIPTION	BY	APP
V1.0	2026-10-12	PRELIMINARY SCREENING	ENG	PM
V3.0	**2026-10-15**	**V3 RIGOROUS FEED & AUDIT**	**SYS**	**DIR**

HARAKUTIR INDUSTRIAL PULSE PROCESSING

15,000 kg/day Incoming-Feed Toor-Dal Tenderization Facility

CONCEPTUAL FRONT-END ENGINEERING DESIGN (FEED) PACKAGE — REV V3.0

PROJECT DESIGN PARAMETERS

- Raw Material: Imported Tanzanian Split Pigeon Pea (Cajanus cajan)
- **Consignment Tracking: PENDING_COMMERCIAL_LOT_ID**
- Incoming Capacity: 15,000 kg/day (13,425 kg/day Dry Solids @ 10.5% wb)
- Finished Dal Yield: 14,267.8 kg/day (95.65% Dry Solids Recovery)
- Chemical Treatment: 0.50% w/w Food-Grade NaHCO3 (112.5 kg/day peak)
- Drying Duty: 3,877.8 kg/day Evaporation (285.1 kg/h Design over 13.6h)
- Turnkey Plant CAPEX: ₹542.0 Lakh (AACE Class 4 Greenfield Estimate)
- Conversion OpEx: ₹9.35 / kg Finished Dal (Full Factory Conversion)
- Plot Dimensions: 52.0m x 24.0m (1,248.0 m² Pre-Engineered Building)
- Governing Standards: ASME Sec VIII, IS 2825, FSSAI 2011, NFPA 68, ISA-5.1

STATUS: GATE G-A HOLD ACTIVE (Zero physical data fabricated)

MASTER DRAWING INDEX

SHEET 01 | DWG-FIN-15T-000 : Master Cover Sheet & Drawing Index	PROJECT
SHEET 02 | DWG-FIN-15T-001 : Greenfield Site Plot Plan & Hygienic Zoning	CIVIL / PLOT
SHEET 03 | DWG-FIN-15T-002 : Process Flow Diagram (PFD) & 12-Stream Balance	PROCESS
SHEET 04 | DWG-FIN-15T-003 : P&ID: Raw Grain Intake & Soaking Battery	PIPING & INST
SHEET 05 | DWG-FIN-15T-004 : P&ID: Vibrating Fluid Bed Drying & Thermal Air	PIPING & INST
SHEET 06 | DWG-FIN-15T-005 : P&ID: Cleanroom Packaging & N2 Flushing	PIPING & INST
SHEET 07 | DWG-FIN-15T-006 : Equipment GA: 7.5 m³ Sanitary Soak Vessel TK-101	MECHANICAL
SHEET 08 | DWG-FIN-15T-007 : Equipment GA: Vibrating Fluid Bed Dryer DR-101	MECHANICAL
SHEET 09 | DWG-FIN-15T-008 : Equipment GA: Automated Chemical Skid SK-100	MECHANICAL
SHEET 10 | DWG-FIN-15T-009 : Electrical Single Line Diagram (SLD) & MCC	ELECTRICAL
SHEET 11 | DWG-FIN-15T-010 : Utility Flow Diagram & 25 KLD ETP Flowsheet	ENVIRONMENTAL
SHEET 12 | DWG-FIN-15T-011 : Hazardous Area Classification & Dust Safety Plan	SAFETY / EHS

GENERAL ENGINEERING NOTES:
1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS SPECIFIED OTHERWISE. LEVEL DATUM ±0.000 CORRESPONDS TO FINISHED FLOOR LEVEL (FFL).
2. THIS DRAWING IS THE PROPERTY OF HARAKUTIR AND CONSTITUTES PRELIMINARY FRONT-END ENGINEERING DESIGN (FEED).
3. ALL EQUIPMENT CAPACITIES, NOZZLE SCHEDULES, AND UTILITIES ARE BASED ON SENSITIVITY ENVELOPES PENDING GATE G-A PHYSICAL TRIAL RESULTS.
4. STRICTLY PROHIBITED FROM USE FOR TENDER, PURCHASE ORDER PROCUREMENT, SHOP FABRICATION, OR SITE CONSTRUCTION.

HARAKUTIR AGRO-PROCESS ENGINEERING

15,000 kg/day Toor-Dal Optimization Greenfield Plant

MASTER COVER SHEET & DRAWING INDEX

DWG NO: DWG-FIN-15T-000

UNITS: MILLIMETERS (mm)

STAGE: CONCEPTUAL FEED

DISCIPLINE: PROJECT MANAGEMENT

STAMP: PRELIMINARY - NOT FOR CONSTRUCTION

REV: V3.0

DATE: OCT 2026

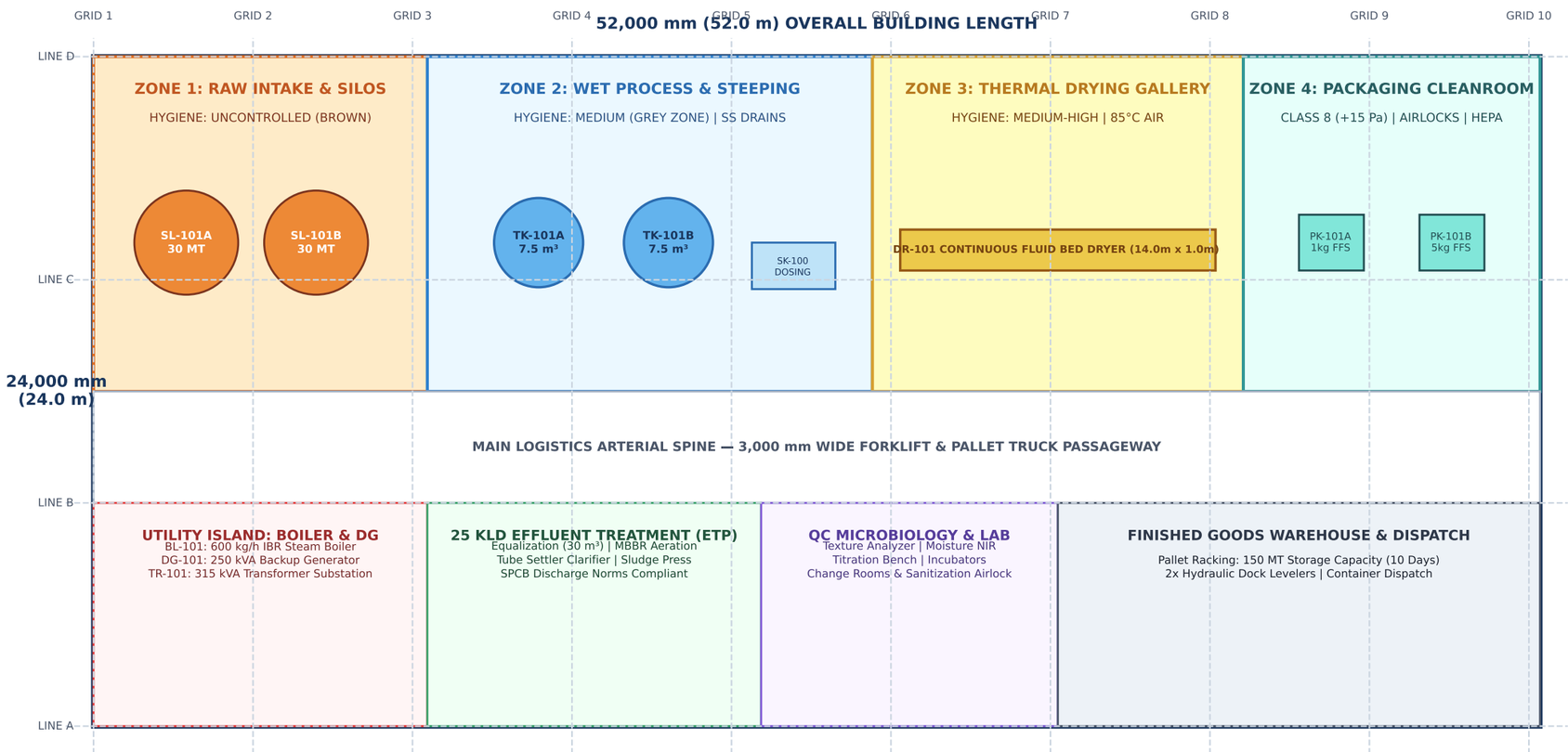
PROJ ID: FIN-15T-2026

SHEET: 01 OF 12

CHK: RED-TEAM

GATE: G-A HOLD

REV	DATE	DESCRIPTION	BY	APP
V1.0	2026-10-12	PRELIMINARY SCREENING	ENG	PM
V3.0	**2026-10-15**	**V3 RIGOROUS FEED & AUDIT**	**SYS**	**DIR**



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HARAKUTIR AGRO-PROCESS ENGINEERING

15,000 kg/day Toor-Dal Optimization Greenfield Plant

GREENFIELD SITE PLOT PLAN & HYGIENIC ZONING

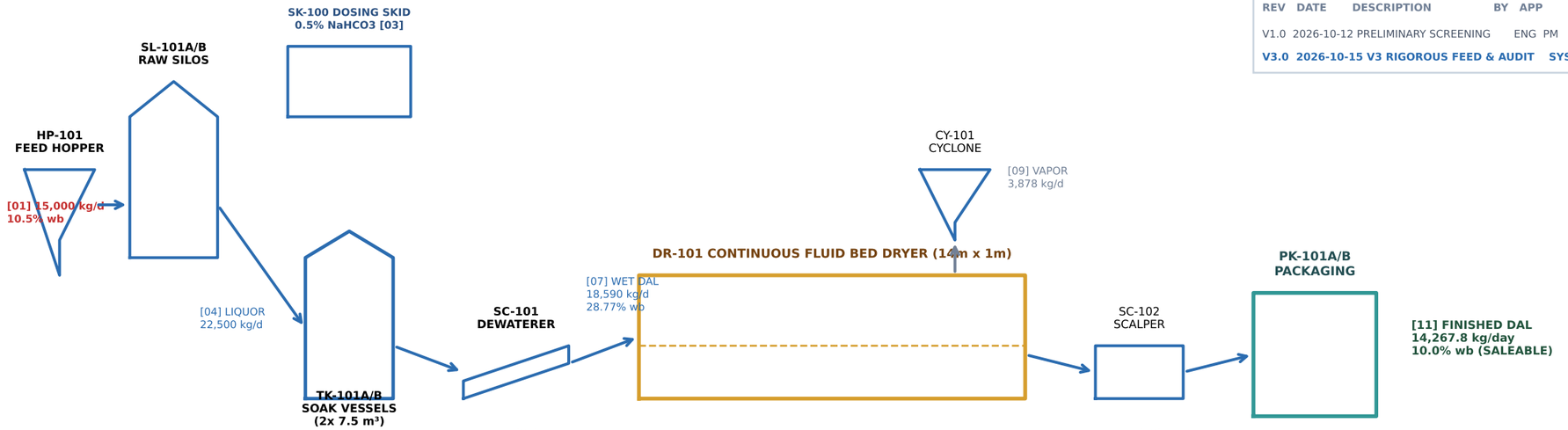
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DWG NO: DWG-FIN-15T-001

UNITS: MILLIMETERS (mm)

STAGE: CONCEPTUAL FEED	**REV: V3.0**	**SHEET: 02 OF 12**
DISCIPLINE: CIVIL / PLOT	DATE: OCT 2026	CHK: RED-TEAM
STAMP: PRELIMINARY - NOT FOR CONSTRUCTION	PROJ ID: FIN-15T-2026	**GATE: G-A HOLD**

REV	DATE	DESCRIPTION	BY	APP
V1.0	2026-10-12	PRELIMINARY SCREENING	ENG	PM
V3.0	**2026-10-15**	**V3 RIGOROUS FEED & AUDIT**	**SYS**	**DIR**



HEAT AND MATERIAL BALANCE TABLE (RIGOROUS CONSERVATION CLOSURE)

STR	STREAM DESCRIPTION	PHASE	TOTAL (kg/d)	DRY SOLIDS	WATER	MOISTURE	SOLUTE	TEMP	PRESS
01	Raw Toor-Dal Intake	Granular Solid	15,000.00	13,425.00	1,575.00	10.50% wb	0.00	30.0 °C	1.0 atm
02	Fresh Process Water Supply	Liquid Aqueous	22,387.50	0.00	22,387.50	100.00% wb	0.00	28.0 °C	3.0 bar g
03	Chemical Dosing (NaHCO3)	Solid Solute	112.50	112.50	0.00	0.00% wb	112.50	30.0 °C	Atm
04	Fresh Soak Liquor (0.50% w/w)	Aqueous Soln	22,500.00	112.50	22,387.50	99.50% wb	112.50	30.0 °C	1.5 bar g
05	Total Soak Tank Charge Slurry	Solid/Liquid Slurry	37,500.00	13,537.50	23,962.50	63.90% wb	112.50	30.0 °C	Atm
06	Spent Soak Effluent to ETP	Wastewater	18,910.00	296.50	18,613.50	98.43% wb	102.50	32.0 °C	Gravity
07	Dewatered Wet Dal to Dryer	Moist Granular	18,590.00	13,241.00	5,349.00	28.77% wb	10.00	30.0 °C	Gravity
08	Dried Dal Discharge from DR-101	Dry Granular	14,712.22	13,241.00	1,471.22	10.00% wb	10.00	55.0 °C	Atm
09	Dryer Exhaust Water Vapor	Vapor / Air	3,877.78	0.00	3,877.78	100.00% wb	0.00	60.0 °C	-50 mmWC
10	Scalper & Deduster Fines Loss	Dust / Broken	444.47	400.03	44.44	10.00% wb	0.30	35.0 °C	Gravity
11	Saleable Finished Packed Dal	Food Grade Dal	14,267.75	12,840.98	1,426.77	10.00% wb	9.70	30.0 °C	Atm
12	CIP Washdown & Floor Drainage	Effluent Wash	4,500.00	20.00	4,480.00	99.56% wb	15.00	45.0 °C	Floor Trench

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3. ALL EQUIPMENT CAPACITIES, NOZZLE SCHEDULES, AND UTILITIES ARE BASED ON SENSITIVITY ENVELOPES PENDING GATE G-A PHYSICAL TRIAL RESULTS.
4. STRICTLY PROHIBITED FROM USE FOR TENDER, PURCHASE ORDER PROCUREMENT, SHOP FABRICATION, OR SITE CONSTRUCTION.

HARAKUTIR AGRO-PROCESS ENGINEERING
15,000 kg/day Toor-Dal Optimization Greenfield Plant

PROCESS FLOW DIAGRAM & HEAT/MATERIAL BALANCE

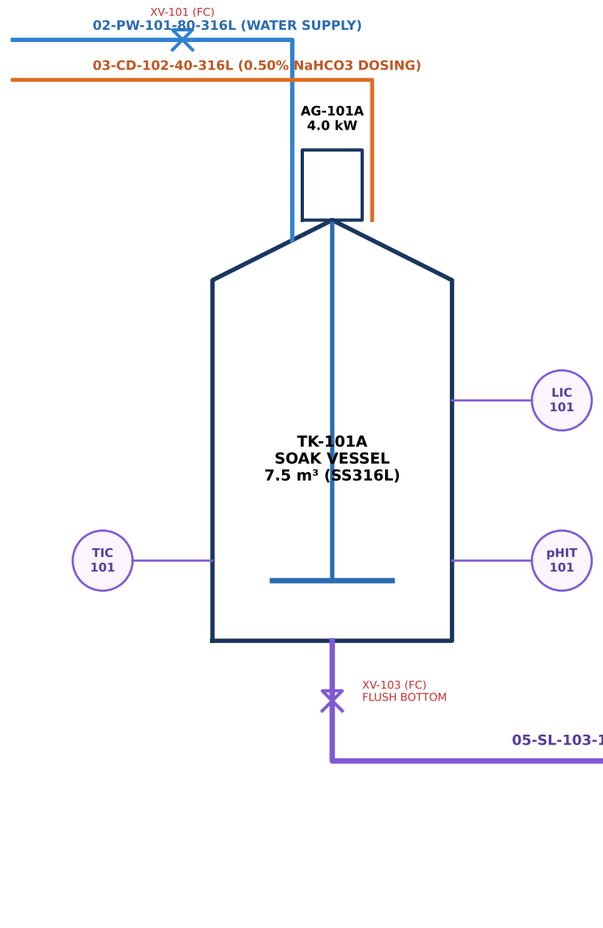
DWG NO: DWG-FIN-15T-002 | SCALE: NTS | UNITS: MILLIMETERS (mm)

STAGE: CONCEPTUAL FEED | **REV: V3.0** | **SHEET: 03 OF 12**

DISCIPLINE: PROCESS ENGINEERING | DATE: OCT 2026 | CHK: RED-TEAM

STAMP: PRELIMINARY - NOT FOR CONSTRUCTION | PROJID: FIN-15T-2026 | **GATE: G-A HOLD**

REV	DATE	DESCRIPTION	BY	APP
V1.0	2026-10-12	PRELIMINARY SCREENING	ENG	PM
V3.0	**2026-10-15**	**V3 RIGOROUS FEED & AUDIT**	**SYS**	**DIR**



SAFETY INTERLOCKS & CONTROL PHILOSOPHY (ISA-84 / SIL-1)

[I-01] High-High Level LAHH-101 activates: Closes Water Feed XV-101 and Dosing XV-102 immediately.

[I-02] Agitator Fault PAL-101: Tripping of AG-101 motor halts dry feed intake conveyor CV-101.

[I-03] Low Level LALL-101: Disables Flush Bottom Drain XV-103 and slurry pump P-101 to prevent dry running.

[I-04] pH Deviation AAL-101: If pH < 7.5 or pH > 8.8, triggers alarm on SCADA and pauses automated batch start.

[I-05] Dewatering Jam ZSL-101: Motor under-current stops tank discharge XV-103 within 3 seconds.

[I-06] Emergency Stop PB-101: Slams all air-actuated valves to fail-safe positions (FC: Fail Closed).

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HARAKUTIR AGRO-PROCESS ENGINEERING

15,000 kg/day Toor-Dal Optimization Greenfield Plant

PIPING & INSTRUMENTATION: SOAKING BATTERY

DWG NO: DWG-FIN-15T-003

SCALE: NTS
UNITS: MILLIMETERS (mm)

STAGE: CONCEPTUAL FEED
DISCIPLINE: CONTROL & PIPING
STAMP: PRELIMINARY - NOT FOR CONSTRUCTION

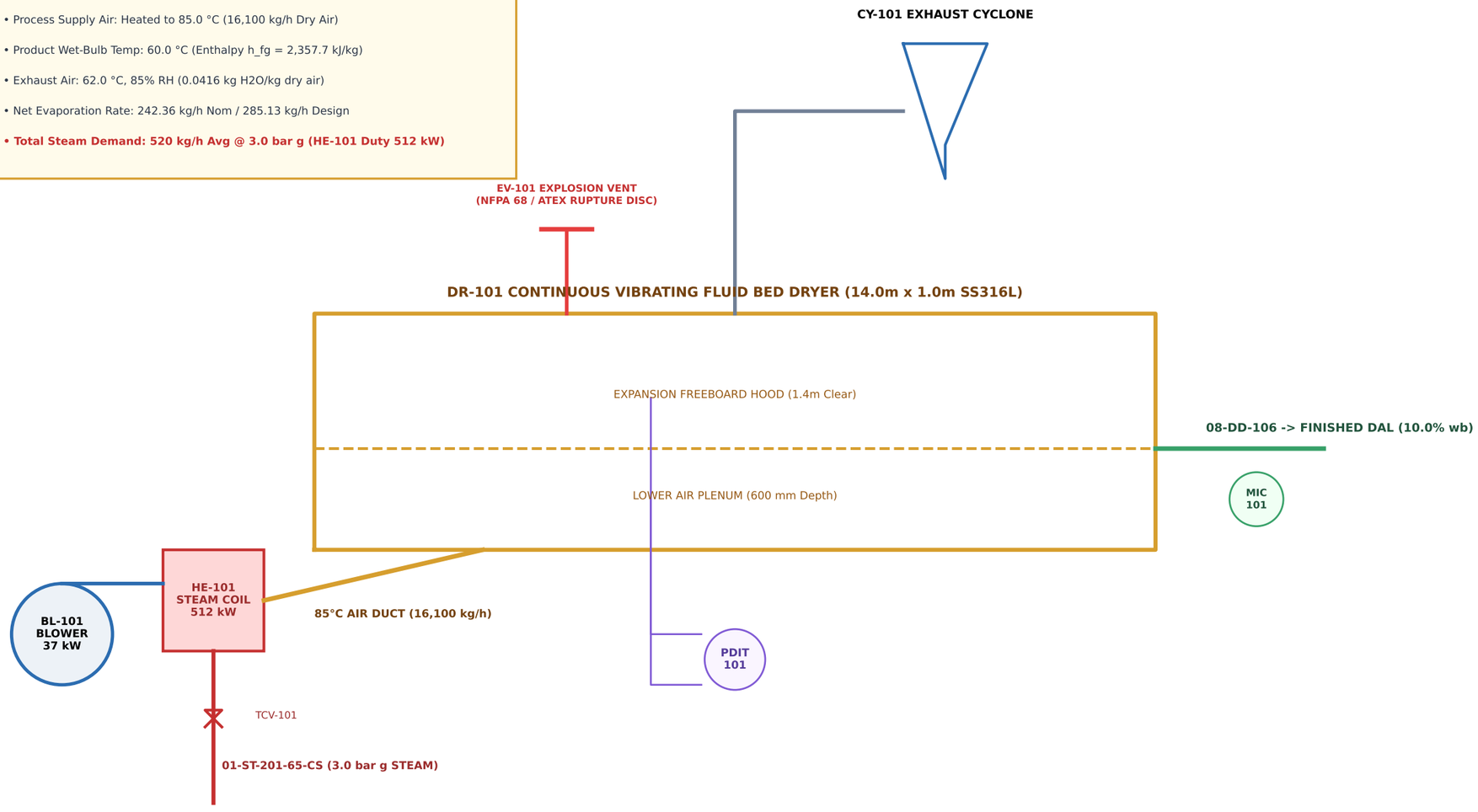
REV: V3.0
DATE: OCT 2026
PROJ ID: FIN-15T-2026

SHEET: 04 OF 12
CHK: RED-TEAM
GATE: G-A HOLD

REV	DATE	DESCRIPTION	BY	APP
V1.0	2026-10-12	PRELIMINARY SCREENING	ENG	PM
V3.0	**2026-10-15**	**V3 RIGOROUS FEED & AUDIT**	**SYS**	**DIR**

DRYER THERMAL & PSYCHROMETRIC DESIGN BASIS

- Ambient Air Intake: 30.0 °C, 65.0% RH (0.0175 kg H2O/kg dry air)
- Process Supply Air: Heated to 85.0 °C (16,100 kg/h Dry Air)
- Product Wet-Bulb Temp: 60.0 °C (Enthalpy h_fg = 2,357.7 kJ/kg)
- Exhaust Air: 62.0 °C, 85% RH (0.0416 kg H2O/kg dry air)
- Net Evaporation Rate: 242.36 kg/h Nom / 285.13 kg/h Design
- **Total Steam Demand: 520 kg/h Avg @ 3.0 bar g (HE-101 Duty 512 kW)**



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HARAKUTIR AGRO-PROCESS ENGINEERING

15,000 kg/day Toor-Dal Optimization Greenfield Plant

PIPING & INSTRUMENTATION: FLUID BED DRYER

DWG NO: DWG-FIN-15T-004

SCALE: NTS

UNITS: MILLIMETERS (mm)

STAGE: CONCEPTUAL FEED

DISCIPLINE: PROCESS & THERMAL

STAMP: PRELIMINARY - NOT FOR CONSTRUCTION

REV: V3.0

DATE: OCT 2026

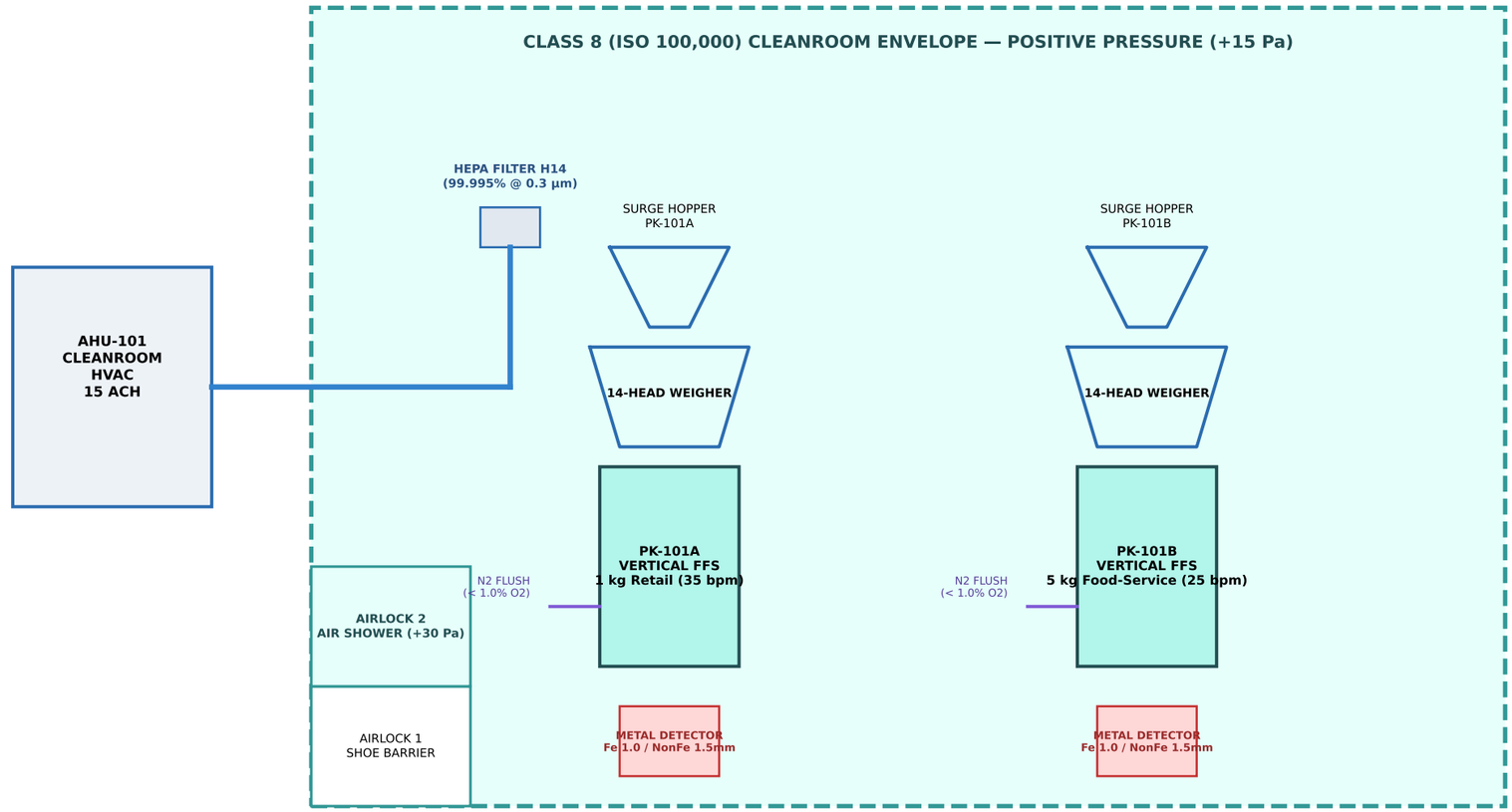
PROJID: FIN-15T-2026

SHEET: 05 OF 12

CHK: RED-TEAM

GATE: G-A HOLD

REV	DATE	DESCRIPTION	BY	APP
V1.0	2026-10-12	PRELIMINARY SCREENING	ENG	PM
V3.0	**2026-10-15**	**V3 RIGOROUS FEED & AUDIT**	**SYS**	**DIR**



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HARAKUTIR AGRO-PROCESS ENGINEERING

15,000 kg/day Toor-Dal Optimization Greenfield Plant

PIPING & INSTRUMENTATION: PACKAGING & HVAC

DWG NO: DWG-FIN-15T-005

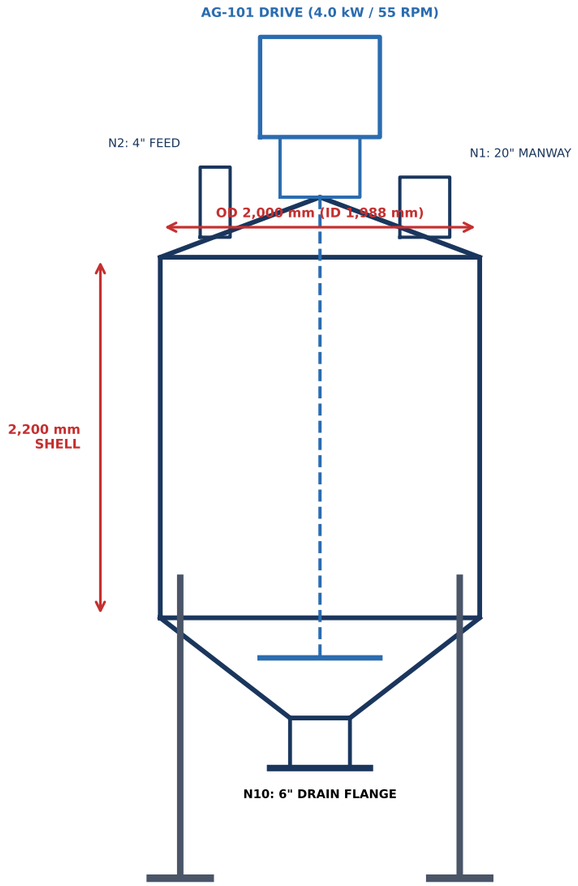
SCALE: NTS
UNITS: MILLIMETERS (mm)

STAGE: CONCEPTUAL FEED
DISCIPLINE: PACKAGING & HVAC
STAMP: PRELIMINARY - NOT FOR CONSTRUCTION

REV: V3.0
DATE: OCT 2026
PROJID: FIN-15T-2026

SHEET: 06 OF 12
CHK: RED-TEAM
GATE: G-A HOLD

REV	DATE	DESCRIPTION	BY	APP
V1.0	2026-10-12	PRELIMINARY SCREENING	ENG	PM
V3.0	2026-10-15	V3 RIGOROUS FEED & AUDIT	SYS	DIR



NOZZLE SCHEDULE & MECHANICAL DESIGN DATA

NOZ	SERVICE	SIZE	RATING	FACING	ORIENTATION
N1	Manway with Sight Glass	20" (500mm)	150#	Sanitary Davit	Top Head 45°
N2	Dal Grain Feed Inlet	4" (100mm)	150#	ANSI RF / TC	Top Head 180°
N3	Softened Water Supply	3" (80mm)	150#	ANSI RF	Top Head 270°
N4	Chemical Dosing Liquor	1.5" (40mm)	150#	ANSI RF	Top Head 90°
N5	CIP Rotary Spray Ball	2" (50mm)	Tri-Clamp	Sanitary 3A	Top Head Center
N6	Sterile Air Breathing Vent	3" (80mm)	Tri-Clamp	0.22 µm PTFE	Top Head 135°
N7	Pressure Relief Valve	2" (50mm)	150#	ANSI RF	Top Head 225°
N8	Guided Wave Radar Level	2" (50mm)	150#	ANSI RF	Top Head 315°
N9	PT100 Duplex Thermowell	1" (25mm)	150#	ANSI RF	Lower Shell 90°
N10	Conical Bottom Slurry Drain	6" (150mm)	150#	Flush Diaphragm	Bottom Center 0°
N11	Vessel Overflow to ETP	3" (80mm)	150#	ANSI RF	Top Shell 0°

MECHANICAL SPECIFICATIONS (ASME SEC VIII DIV 1 / IS 2825)

- Vessel Gross Capacity: 6.83 m³ Fluid (7.50 m³ Envelope) | Working: 5.12 m³
- Design Pressure: Full Vacuum to 1.5 bar g | Test Pressure: 2.25 bar g (Hydro)
- Design Temperature: 100.0 °C | Operating Temperature: 25.0 °C to 45.0 °C
- Material of Construction: SS316L Contact Parts (Ra < 0.8 µm Electro-polished)
- Shell Thickness: 6.0 mm | Head Thickness: 6.0 mm | Cone Thickness: 6.0 mm
- **Empty Tare Weight: 809.0 kg SS316L | Operating Weight (Full Slurry): 8,309 kg**
- Hydrotest Weight: 9,250 kg | Seismic Zone: Zone III | Wind Load: 44 m/s

STAMP: CONCEPTUAL FEED GA — PRELIMINARY — NOT FOR FABRICATION

GENERAL ENGINEERING NOTES:
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HARAKUTIR AGRO-PROCESS ENGINEERING

15,000 kg/day Toor-Dal Optimization Greenfield Plant

EQUIPMENT GA: 7.5 M³ SOAK TANK TK-101

DWG NO: DWG-FIN-15T-006

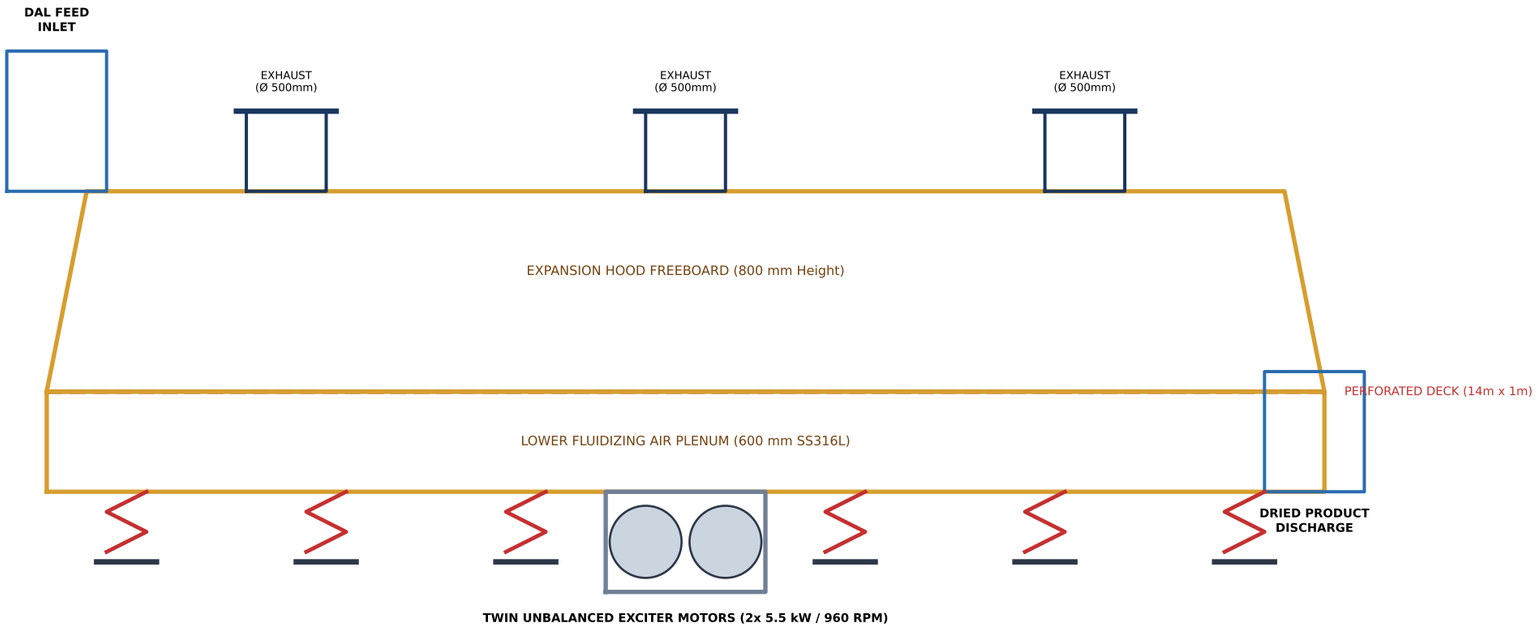
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UNITS: MILLIMETERS (mm)

STAGE: CONCEPTUAL FEED
DISCIPLINE: MECHANICAL / VESSEL
STAMP: PRELIMINARY - NOT FOR CONSTRUCTION

REV: V3.0
DATE: OCT 2026
PROJ ID: FIN-15T-2026

SHEET: 07 OF 12
CHK: RED-TEAM
GATE: G-A HOLD

REV	DATE	DESCRIPTION	BY	APP
V1.0	2026-10-12	PRELIMINARY SCREENING	ENG	PM
V3.0	**2026-10-15**	**V3 RIGOROUS FEED & AUDIT**	**SYS**	**DIR**



CIVIL FOUNDATION LOADING & DYNAMIC SPECIFICATIONS

STATION ID	LOCATION (X)	STATIC DEAD LOAD	GRAIN LIVE LOAD	DYNAMIC AMPLITUDE	SPRING STIFFNESS	TRANSMISSIBILITY
ISOLATOR 1 & 2	X = 1,200 mm (Feed End)	850 kg / station	420 kg / station	± 3.2 mm @ 16 Hz	185 N/mm	< 4.5%
ISOLATOR 3 & 4	X = 3,400 mm	850 kg / station	420 kg / station	± 3.5 mm @ 16 Hz	185 N/mm	< 4.5%
ISOLATOR 5 & 6	X = 5,600 mm (Exciter Mid)	1,250 kg / station	450 kg / station	± 4.0 mm @ 16 Hz	240 N/mm	< 4.0%
ISOLATOR 7 & 8	X = 8,400 mm	1,250 kg / station	450 kg / station	± 4.0 mm @ 16 Hz	240 N/mm	< 4.0%
ISOLATOR 9 & 10	X = 10,600 mm	850 kg / station	420 kg / station	± 3.5 mm @ 16 Hz	185 N/mm	< 4.5%
ISOLATOR 11 & 12	X = 12,800 mm (Discharge)	850 kg / station	420 kg / station	± 3.2 mm @ 16 Hz	185 N/mm	< 4.5%

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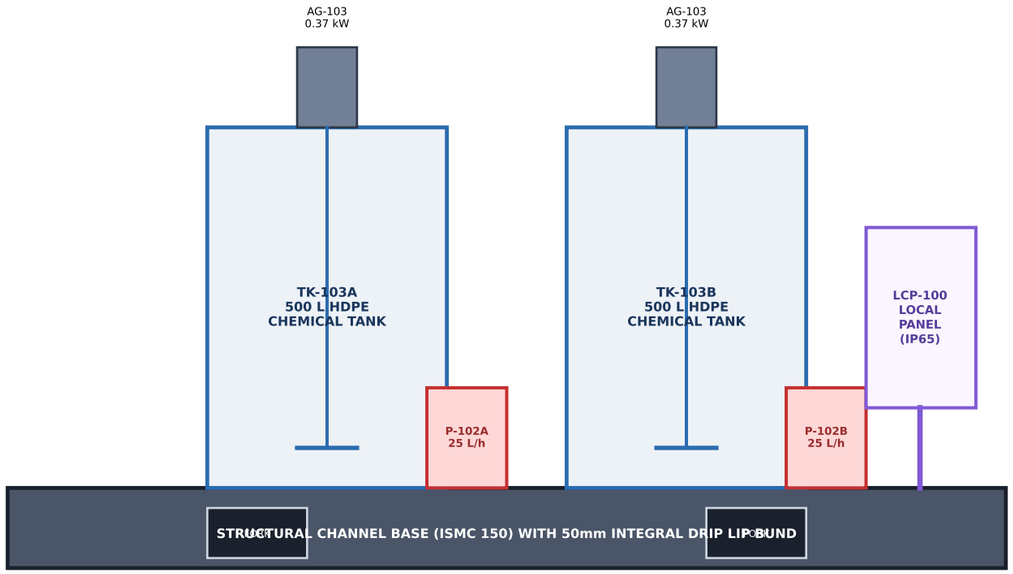
HARAKUTIR AGRO-PROCESS ENGINEERING

15,000 kg/day Toor-Dal Optimization Greenfield Plant

EQUIPMENT GA: FLUID BED DRYER DB-101

DWG NO: DWG-FIN-15T-007	UNITS: MILLIMETERS (mm)	
STAGE: CONCEPTUAL FEED	**REV: V3.0**	**SHEET: 08 OF 12**
DISCIPLINE: MECHANICAL / THERMAL	DATE: OCT 2026	CHK: RED-TEAM
STAMP: PRELIMINARY - NOT FOR CONSTRUCTION	PROJID: FIN-15T-2026	**GATE: G-A HOLD**

REV	DATE	DESCRIPTION	BY	APP
V1.0	2026-10-12	PRELIMINARY SCREENING	ENG	PM
V3.0	**2026-10-15**	**V3 RIGOROUS FEED & AUDIT**	**SYS**	**DIR**



SKID BILL OF MATERIALS (BOM)

ITEM	DESCRIPTION	QTY	MATERIAL	SPEC
01	Dosing Storage Tank	2 No	HDPE UV	500 L Graduated
02	Top Marine Propeller Mixer	2 No	SS316L	0.37 kW / 280 RPM
03	Diaphragm Metering Pump	2 No	PTFE / PVDF	25.0 L/h @ 5 bar
04	Calibration Glass Cylinder	1 No	Borosilicate	1,000 mL with 3-way
05	Pulsation Dampener	1 No	SS316L/Viton	0.5 L @ 60% P_op
06	Back Pressure Valve	2 No	PTFE / Hast-C	Set @ 1.5 bar g
07	Pressure Relief Valve	2 No	SS316L	Set @ 3.5 bar g
08	Suction Y-Strainer	2 No	SS316L	40 Mesh Basket
09	Magnetic Flowmeter FIT	1 No	PFA Lined	DN15, 4-20mA HART
10	Pressure Transmitter PIT	2 No	SS316L Diaph	0-10 bar g IP67
11	Structural Steel Frame	1 Lot	ISMC 150 CS	Hot-Dip Galv 80µm
12	Local Control Panel LCP	1 No	SS304 IP65	VFD + HOA Switches

GENERAL ENGINEERING NOTES:
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HARAKUTIR AGRO-PROCESS ENGINEERING

15,000 kg/day Toor-Dal Optimization Greenfield Plant

EQUIPMENT GA: DOSING SKID SK-100

DWG NO: DWG-FIN-15T-008

SCALE: 1:20

UNITS: MILLIMETERS (mm)

STAGE: CONCEPTUAL FEED

DISCIPLINE: MECHANICAL / SKID

STAMP: PRELIMINARY - NOT FOR CONSTRUCTION

REV: V3.0

DATE: OCT 2026

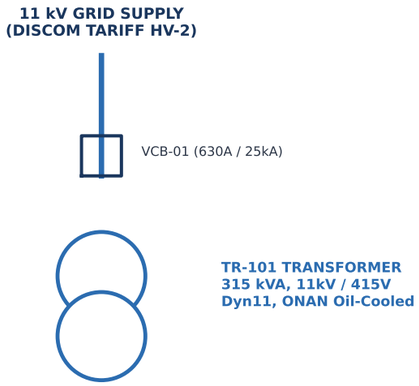
PROJ: FIN-15T-2026

SHEET: 09 OF 12

CHK: RED-TEAM

GATE: G-A HOLD

REV	DATE	DESCRIPTION	BY	APP
V1.0	2026-10-12	PRELIMINARY SCREENING	ENG	PM
V3.0	2026-10-15	V3 RIGOROUS FEED & AUDIT	SYS	DIR



MOTOR CONTROL CENTER (MCC) CONNECTED LOAD SCHEDULE

FEEDER ID	LOAD DESCRIPTION	MOTOR TAG	QTY	RATING (kW)	STARTING METHOD	FULL LOAD AMPS	BREAKER (MCCB)	CABLE SIZE
F-01	Soak Vessel Agitators	AG-101A/B	2 No	4.0 kW ea	VFD (Variable Speed)	8.2 A	16A 3P MCCB	4C x 4 mm² Cu
F-02	Fluid Bed Dryer Fluidizing Blower	BL-101	1 No	37.0 kW	VFD (Inverter Duty)	68.0 A	100A 3P MCCB	3C x 70 mm² Al
F-03	Dryer Exhaust Induced Draft Fan	BL-102	1 No	18.5 kW	Star-Delta Starter	34.5 A	50A 3P MCCB	3C x 25 mm² Al
F-04	Dryer Vibratory Exciter Motors	M-101A/B	2 No	5.5 kW ea	Direct-On-Line (DOL)	11.2 A	25A 3P MCCB	4C x 6 mm² Cu
F-05	Dewatering Vibratory Screen	M-102	1 No	3.7 kW	Direct-On-Line (DOL)	7.8 A	16A 3P MCCB	4C x 4 mm² Cu
F-06	Chemical Dosing Metering Pumps	P-102A/B	2 No	0.75 kW ea	VFD Micro-Drive	2.1 A	6A 3P MCB	4C x 2.5 mm² Cu
F-07	Package Steam Boiler Feed Pumps	P-103A/B	2 No	5.5 kW ea	Direct-On-Line (DOL)	11.2 A	25A 3P MCCB	4C x 6 mm² Cu
F-08	Cleanroom Packaging Lines FFS	PK-101A/B	2 No	12.0 kW ea	Servo & Inverter	24.5 A	40A 3P MCCB	4C x 16 mm² Cu
F-09	Cleanroom HVAC AHU Supply Fan	AHU-101	1 No	15.0 kW	VFD Pressure Control	28.0 A	50A 3P MCCB	3C x 25 mm² Al
F-10	Plant Screw Air Compressor	CP-101	1 No	22.0 kW	VFD Energy Saver	41.0 A	63A 3P MCCB	3C x 35 mm² Al
F-11	25 KLD ETP Aeration Blowers	ET-BL-01	2 No	7.5 kW ea	Star-Delta Starter	15.0 A	32A 3P MCCB	4C x 10 mm² Cu
F-12	Plant Base Lighting & Automation	LDB / SCADA	1 Lot	15.0 kW tot	Feeder Contactor	26.0 A	40A 4P MCB	4C x 16 mm² Cu

TOTAL CONNECTED PLANT LOAD: 162.2 kW | MAXIMUM DEMAND (@ 0.75 DIVERSITY): 121.6 kW | BASE RUNNING LOAD: 85.0 kW

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HARAKUTIR AGRO-PROCESS ENGINEERING

15,000 kg/day Toor-Dal Optimization Greenfield Plant

ELECTRICAL SINGLE LINE DIAGRAM (SLD)

DWG NO: DWG-FIN-15T-009

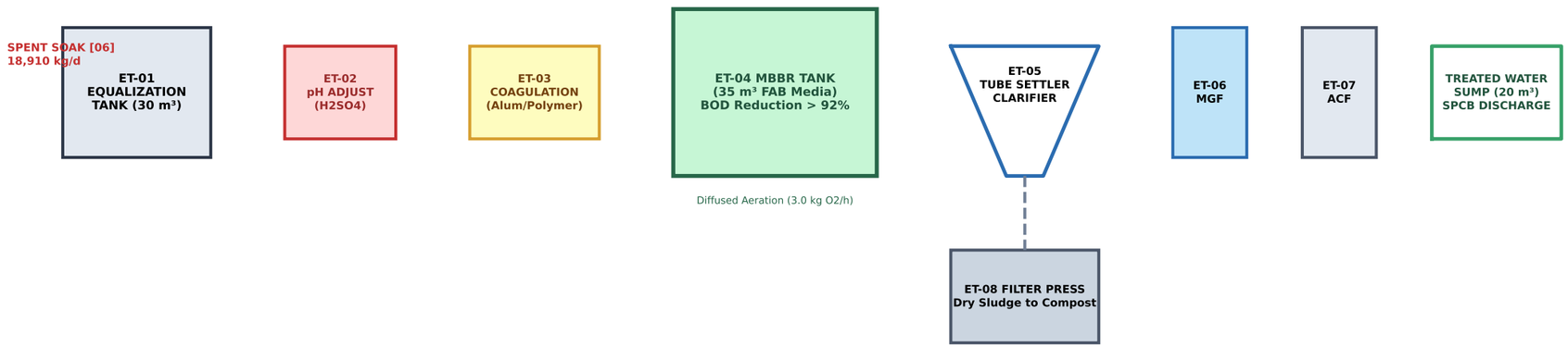
SCALE: NTS
UNITS: MILLIMETERS (mm)

STAGE: CONCEPTUAL FEED
DISCIPLINE: ELECTRICAL / AUTOMATION
STAMP: PRELIMINARY - NOT FOR CONSTRUCTION

REV: V3.0
DATE: OCT 2026
PROJID: FIN-15T-2026

SHEET: 10 OF 12
CHK: RED-TEAM
GATE: G-A HOLD

REV	DATE	DESCRIPTION	BY	APP
V1.0	2026-10-12	PRELIMINARY SCREENING	ENG	PM
V3.0	**2026-10-15**	**V3 RIGOROUS FEED & AUDIT**	**SYS**	**DIR**



UTILITY & ENVIRONMENTAL QUALITY PARAMETERS TABLE

PARAMETER / STREAM	RAW BOREWELL	SOFTENED WATER	BOILER FEEDWATER	RAW EFFLUENT	TREATED ETP EFFLUENT	SPCB LIMIT
pH Value	7.4 - 7.8	7.2 - 7.6	8.5 - 9.5 (with morph)	8.2 - 8.9 (alkaline soak)	7.0 - 7.5	6.5 - 8.5
Total Dissolved Solids (TDS)	650 mg/L	680 mg/L	< 1,500 mg/L (blowdown)	1,850 mg/L	1,200 mg/L	< 2,100 mg/L
Total Hardness (as CaCO3)	320 mg/L	< 5 mg/L (Zero Hard)	< 1 mg/L	280 mg/L	< 50 mg/L	N/A
Chemical Oxygen Demand (COD)	< 10 mg/L	< 5 mg/L	< 5 mg/L	2,400 mg/L (leached)	< 180 mg/L	< 250 mg/L
Biochemical Oxygen Demand (BOD5)	< 3 mg/L	< 2 mg/L	< 2 mg/L	1,250 mg/L	< 25 mg/L	< 30 mg/L
Total Suspended Solids (TSS)	< 15 mg/L	< 2 mg/L	< 1 mg/L	480 mg/L (fines)	< 30 mg/L	< 100 mg/L
Oil & Grease	Nil	Nil	Nil	35 mg/L	< 5 mg/L	< 10 mg/L
Dissolved Oxygen (DO)	5.5 mg/L	6.0 mg/L	< 0.02 mg/L (deaerated)	< 0.5 mg/L	> 4.0 mg/L	> 4.0 mg/L

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HARAKUTIR AGRO-PROCESS ENGINEERING

15,000 kg/day Toor-Dal Optimization Greenfield Plant

UTILITY FLOW DIAGRAM & 25 KLD ETP

SCALE: NTS

DWG NO: DWG-FIN-15T-010

UNITS: MILLIMETERS (mm)

STAGE: CONCEPTUAL FEED

DISCIPLINE: ENVIRONMENTAL & UTILITY

STAMP: PRELIMINARY - NOT FOR CONSTRUCTION

REV: V3.0

DATE: OCT 2026

PROJID: FIN-15T-2026

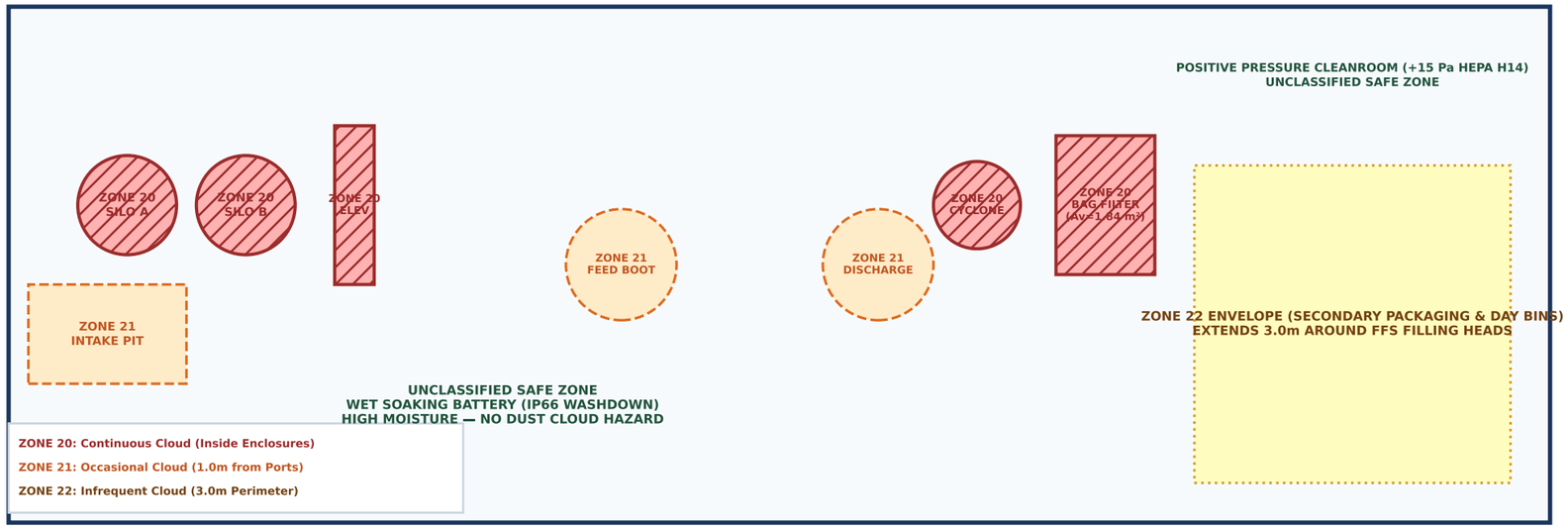
SHEET: 11 OF 12

CHK: RED-TEAM

GATE: G-A HOLD

REV	DATE	DESCRIPTION	BY	APP
V1.0	2026-10-12	PRELIMINARY SCREENING	ENG	PM
V3.0	**2026-10-15**	**V3 RIGOROUS FEED & AUDIT**	**SYS**	**DIR**

MAIN FACILITY DUST HAZARD CONTOUR MAP (NFPA 61 / NFPA 652 / NFPA 68 / ATEX 2014/34/EU)



NFPA 68 / NFPA 652 EXPLOSION VENTING DESIGN BASIS

• **Combustible Dust Specimen:**	Tanzanian Split Pigeon Pea (Toor-Dal) Flour / Fines
• **Median Particle Size (D50):**	42 µm (< 75 µm sieve fraction passing > 92%)
• **Dust Explosibility Class:**	St 1 (Low to Moderate Deflagration Severity)
• **Maximum Explosion Pressure (Pmax):**	8.5 barg (850 kPa gauge)
• **Deflagration Index (Kst):**	110 bar·m/s (measured in 20L sphere per ASTM E1226)
• **Minimum Ignition Energy (MIE):**	45 mJ (capacitive spark without inductance)
• **Minimum Ignition Temp - Cloud (MIT):**	420 °C (BAM furnace per ASTM E1491)
• **Minimum Ignition Temp - 5mm Layer:**	290 °C (hot plate per ASTM E2021)
• **Maximum Equipment Surface Temp:**	120 °C (Temp Class T4 / T120°C per EN 50281-1-1)
• **Reduced Pressure Limit (Pred):**	0.40 barg (Baghouse Pred design rating = 0.50 barg)
• **Static Activation Pressure (Pstat):**	0.10 barg ± 15% (Bursting Panel)
• **Calculated Baghouse Vent Area (Av):**	1.84 m² (NFPA 68 Eq. 8.2.2 with L/D = 2.1)
• **Relief Duct Discharge Routing:**	Vertical SS304 duct through roof (< 3m length)

EQUIPMENT INGRESS, GROUNDING & SAFETY INTERLOCKS

ELECTRICAL INGRESS RATINGS:

- Zone 20 / 21 Motors & Enclosures
- Zone 22 Motors & Sensors
- Wet Process Area Instruments
- Substation / MCC Room

STANDARDS & CERTIFICATIONS:

- **Ex tb IIIC T120°C Db IP66 (ATEX / PESO Certified)**
- **Ex tc IIIB T120°C Dc IP65 (Dust-Ignition Proof)**
- **IP66 / NEMA 4X Washdown Stainless Steel SS316L**
- **Pressurized Clean Air (+25 Pa, Non-Hazardous Safe)**

STATIC BONDING & GROUNDING:

- Continuous Earth Grid Resistance
- Piping Flange Jumpers
- Flexible Expansion Sleeves

RESISTANCE SPECIFICATIONS:

- **< 1.0 Ohm to main substation grounding ring**
- **Braided copper continuity jumpers (< 10 Ohms)**
- **Conductive silicone with molded copper wire inserts**

SAFETY INSTRUMENTED SYSTEMS:

- Spark Detection in Exhaust Duct
- Bag Filter Broken Bag Detector
- Silo High-High Level Sensor

ACTION & FAIL-SAFE MATRIX:

- **Grecon IR Sensor -> Fast Gate Closure (< 35 ms)**
- **Triboelectric Probe (AIT-102) -> Auto ID Fan Trip**
- **Tuning Fork LSU-101 -> Interlocks Infeed Conveyor**

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HARAKUTIR AGRO-PROCESS ENGINEERING
15,000 kg/day Toor-Dal Optimization Greenfield Plant

HAZARDOUS AREA CLASSIFICATION & DUST SAFETY PLAN

DWG NO: DWG-FIN-15T-011	SCALE: 1:100	
	UNITS: MILLIMETERS (mm)	
STAGE: CONCEPTUAL FEED	**REV: V3.0**	**SHEET: 12 OF 12**
DISCIPLINE: SAFETY & LOSS PREVENTION	DATE: OCT 2026	CHK: RED-TEAM
STAMP: PRELIMINARY - NOT FOR CONSTRUCTION	PROJ ID: FIN-15T-2026	**GATE: G-A HOLD**